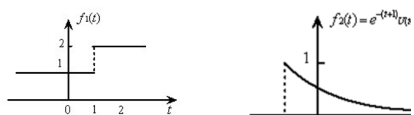


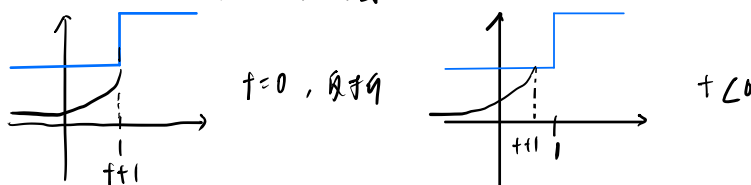
5-1 求 $y(t) = f_1(t) * f_2(t)$, 并画出 $y(t)$ 的波形



$$f_1(t) = 1 \cdot U(t-1)$$

$$f_2(t) = e^{-(t+1)} U(t+1)$$

当 $t \leq 0$ 时, $f_1(t)$ 和 $f_2(t-\tau)$ 的图:



$$y(t) = f_1(t) * f_2(t)$$

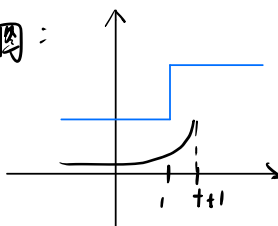
$$= \int_{-\infty}^{t+1} f_1(\tau) \cdot f_2(t-\tau) d\tau$$

$$= \int_{-\infty}^{t+1} 1 \cdot e^{-(t-\tau+1)} d\tau$$

$$\text{设 } u = -t + \tau - 1, \quad \frac{du}{d\tau} = 1 \Rightarrow du = d\tau$$

$$= \int_{-\infty}^0 e^u du = 1$$

当 $t > 0$ 时, $f_1(t)$ 和 $f_2(t-\tau)$ 的图:



$$y(t) = f_1(t) * f_2(t)$$

$$= \int_{-\infty}^{t+1} f_1(\tau) \cdot f_2(t-\tau) d\tau$$

$$= \int_{-\infty}^{t+1} f_1(\tau) \cdot f_2(t-\tau) d\tau$$

$$= \int_{-\infty}^1 f_1(\tau) \cdot f_2(t-\tau) d\tau + \int_1^{t+1} f_1(\tau) \cdot f_2(t-\tau) d\tau$$

$$= \int_{-\infty}^1 1 \cdot e^{-(t-\tau+1)} d\tau + \int_1^{t+1} 2 \cdot e^{-(t-\tau+1)} d\tau$$

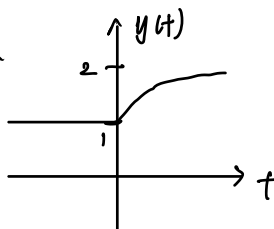
$$\text{设 } u = -(t-\tau+1) = -t + \tau - 1, \quad \frac{du}{d\tau} = 1 \Rightarrow du = d\tau$$

$$= \int_{-\infty}^{-t} e^u du + 2 \cdot \int_{-t}^0 e^u du$$

$$= e^{-t} + 2(1 - e^{-t})$$

$$= 2 - e^{-t}$$

因此 $y(t) = \begin{cases} 1 & t \leq 0 \\ 2e^{-t} & t > 0 \end{cases}$, 波形是



5-2 证明自相关函数的性质:

- 1) $R_x(0)$ 为自相关函数的最大值: $R_x(0) \geq R_x(\tau)$
- 2) 自相关函数的取值范围: $\mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2$
- 3) $R_x(-\tau) = R_x(\tau)$ $R_x(\tau)$ 是 τ 的偶函数

$$1) R_x(\tau) = \left(\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t-\tau) dt \right)^2 \\ = E[x(t) \cdot x(t-\tau)]^2$$

根据柯西-许瓦兹不等式, $E[(x-\mu_x)(y-\mu_y)]^2 \leq E[(x-\mu_x)^2] E[(y-\mu_y)^2]$

$$\text{因此, } R_x(\tau) \leq E[(x(t))^2] E[(x(t-\tau))^2] = E[x^2(t)] E[x^2(t-\tau)] \\ = E[x(t) \cdot x(t-0)] E[x(t) \cdot x(t-0)] \\ = R_x(0) R_x(0) \\ = R_x^2(0)$$

$$\text{因此, } R_x(\tau) \leq R_x^2(0)$$

$$2) \rho_x(\tau) = \frac{E[(x(t)-\mu_x)(x(t-\tau)-\mu_x)]}{\sigma_x \sigma_x} = \frac{R_x(\tau) - \mu_x^2}{\sigma_x^2}$$

$$\therefore |\rho_x| \leq 1$$

$$\therefore \left| \frac{R_x(\tau) - \mu_x^2}{\sigma_x^2} \right| \leq 1$$

$$-\sigma_x^2 \leq R_x(\tau) - \mu_x^2 \leq \sigma_x^2$$

$$\mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2 \quad (\text{得证})$$

$$3) R_x(-\tau) = R_x(\tau)$$

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t-\tau) dt = E[x(t) x(t-\tau)]$$

$$\text{令 } t' = t + \tau \Rightarrow t = t' - \tau,$$

$$R_x(-\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t+\tau) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t' - \tau) x(t') dt$$

$$= R_x(\tau)$$

$$\therefore R_x(-\tau) = R_x(\tau) \quad (\text{得证})$$

5-3 证明互相关函数的性质：

1) 互相关函数的取值范围： $(\mu_x \mu_y - \sigma_x \sigma_y) \leq R_{xy}(\tau) \leq (\mu_x \mu_y + \sigma_x \sigma_y)$

2) $R_{xy}(-\tau) = R_{yx}(\tau)$

1) 根据相关函数：

$$\begin{aligned}
 R_{xy}(\tau) &= \frac{E[(x(t) - \mu_x)(y(t-\tau) - \mu_y)]}{\sigma_x \sigma_y} \\
 &= \frac{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (x(t) - \mu_x)(y(t-\tau) - \mu_y) dt}{\sigma_x \sigma_y} \\
 &= \frac{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} [x(t)y(t-\tau) - x(t)\mu_y - y(t-\tau)\mu_x + \mu_x\mu_y] dt}{\sigma_x \sigma_y} \\
 &= \frac{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)y(t-\tau) dt - \mu_y \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) dt - \mu_x \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t-\tau) dt + \mu_x\mu_y}{\sigma_x \sigma_y} \\
 &= \frac{R_{xy}(\tau) - \mu_x\mu_y - \mu_y\mu_x + \mu_x\mu_y}{\sigma_x \sigma_y} \\
 &= \frac{R_{xy}(\tau) - \mu_x\mu_y}{\sigma_x \sigma_y}
 \end{aligned}$$

$$\therefore |R_{xy}(\tau)| \leq 1$$

$$\therefore \left| \frac{R_{xy}(\tau) - \mu_x\mu_y}{\sigma_x \sigma_y} \right| \leq 1$$

$$-\sigma_x \sigma_y \leq R_{xy}(\tau) - \mu_x\mu_y \leq \sigma_x \sigma_y$$

$$\mu_x\mu_y - \sigma_x \sigma_y \leq R_{xy}(\tau) \leq \mu_x\mu_y + \sigma_x \sigma_y$$

2) $R_{xy}(-\tau) = R_{yx}(\tau)$

$$R_{yx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t)x(t-\tau) dt$$

$$\text{设 } t' = t + \tau \Rightarrow t' - \tau = t$$

$$R_{xy}(-\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t')y(t'-\tau) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t+\tau)y(t) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t)x(t-\tau) dt$$

$$= R_{yx}(\tau) \quad (\text{得证})$$